Aryan Mehta

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.80
Abhijat Vidhyavihar	Class XII (Gujarat Secondary and Higher Secondary Education Board)	2023	82.20%
A.G. High School	Class X (Gujarat Secondary and Higher Secondary Education Board)	2021	86.00%

RESEARCH EXPERIENCE

Soccer Commentary Generation

- **Co-authored a survey paper** analyzing the landscape of soccer commentary generation.
- Developed a **multimodal soccer commentary generation system** integrating Broadcast2Pitch GSR, T-DEED event spotting, temporal aggregation, and GPT-5.4 with rolling context to generate temporally grounded, coherent play-by-play commentary from broadcast video.

PROJECTS

Edge Efficient Deepfake Detection

- Designed an efficient video deepfake detection pipeline with frame sampling, face detection, and region-based feature extraction.
- Developed a lightweight **MobileNet + HTSM** detector leveraging hierarchical temporal modelling to capture temporal dependencies.
- Benchmarked **MobileNet + HTSM, XceptionNet** on **FaceForensics++**, achieving **90.2% lower energy consumption than XceptionNet** while maintaining competitive detection performance.

LLM Reasoning and Safety Classification

- Built models to **classify reasoning trajectories** as human vs LLM-generated, leveraging both classical ML **(Random Forest)** and deep architectures **(BERT + BiGRU),** achieving up to **97.5% accuracy** on test data
- Developed a **step-level** and **trajectory-level** safety classification system using **DeBERTa** and rule-based risk signals to detect unsafe reasoning patterns in LLM outputs.
- Engineered **features from reasoning traces** (token counts, LaTeX usage, hedging patterns) and designed multi-stage pipelines combining **NLI scoring + MLP aggregation** for safety prediction

Reinforcement Learning

- Developed **DQN agents for MountainCar-v0 and Pong-v0**, using experience replay and image preprocessing for pixel-based state.
- Developed **Policy Gradient agents for CartPole-v0 and LunarLander-v2**, incorporating reward-to-go and advantage normalization for variance reduction
- Conducted controlled experiments on **exploration strategies, hyperparameters, and batch sizes**, evaluating their effects on learning stability and convergence.

Foundations of Machine Learning

- Built a configurable **K-Nearest Neighbours** (KNN) algorithm with support for multiple distance metrics; tested on MNIST datasets.
- Implemented **Linear Regression** and studied the impact of hyperparameters on noisy data.

- Developed and evaluated **multiple classifiers** including Least Squares, LDA, FDA, Perceptron, and Logistic Regression -- with and without the presence of outliers.
- Built the **Expectation-Maximization (EM)** algorithm for **Gaussian Mixture Models**; analyzed convergence behavior.
- Designed and trained neural networks on MNIST and non-linear datasets using Adam, dropout, early stopping, and regularization.

Robotic Arm: Modelling & Path Planning

- Implemented forward/inverse kinematics and velocity kinematics of **Puma360** using **DH-parameter modelling**.
- Developed linear, arc, parabolic, and RRT **path planning** algorithms.
- Implemented **cubic, quintic, LSPB, and bang-bang trajectory** generation and evaluated trajectories in MuJoCo.

RELEVANT COURSES

- **Artificial Intelligence**: Computer Vision, Natural Language Processing, Topics in Deep learning, Deep Learning, Reinforcement Learning, Foundation of Machine Learning, AI in Finance*, Multi-Agent Systems*

- **Computer Science**: Discrete Mathematics, Data Structures, Algorithms, Computer Architecture, Database Management.

SKILLS

Languages & Tools: Python, C, C++, HTML, CSS, Bash, Git, SQL, LaTeX

Libraries & Frameworks: NumPy, Scikit-learn, Pytorch, Open AI Gym, OpenCV

EXTRACURRICULAR ACHIEVEMENTS

- **3× National Champion** - 2x in U21 Pairs (2024, 2026) and in U16 Pairs and Teams (2019)
- **Represented Team India** at World Youth Bridge Championships in **Croatia (U16, 2019), Italy (U21, 2025), and China (U21, 2026)**; played as **Captain** in 2026.